Exponents and Positive Powers of 10

You can attempt each problem as many times as needed. To check your answers, click on the blue box with a question mark, or use the Enter key on your keyboard. Do not inlcude commas in your answers.

Exponential Notation

Exponential notation is shorthand for repeated multiplication of the same quantity. For example, the product $5 \cdot 5 \cdot 5$ can be rewritten more simply as 5^3. In this example, the number 5 is called the **base** and the number 3 is called the **exponent**. In general, the exponent indicates the number of times to multiply the base.

Question 1 *Identify the base and exponent of* 10^7.

Base $= 10$ ***Exponent*** $= 7$

Question 2 *Identify the base and exponent of* 4^5.

Base $= 4$ ***Exponent*** $= 5$

Question 3 *Identify the base and exponent of* $\left(\frac{1}{7}\right)^2$.

Base $= \frac{1}{7}$ ***Exponent*** $= 2$

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Question 4 *Write the expression* $11 \cdot 11$ *using exponential notation.*

Multiple Choice:

(a) $2 \cdot 11$

(b) 121

(c) 11^2 ✓

Question 5 *Write the expression* $9 \cdot 9 \cdot 9 \cdot 9$ *using exponential notation.*

Multiple Choice:

(a) $9 \cdot 4$

(b) 9^4 ✓

(c) 9^3

Question 6 *Write the expression* 2^5 *as a product without the exponent.*

Multiple Choice:

(a) $2 \cdot 2 \cdot 2 \cdot 2 \cdot 2$ ✓

(b) $2 \cdot 5$

(c) $2 \cdot 2 \cdot 2 \cdot 2$

Question 7 *Write the expression* $\left(\frac{1}{5}\right)^3$ *as a product without the exponent.*

Multiple Choice:

(a) $\left(\frac{1}{5}\right) \cdot 3$

(b) $\left(\frac{1}{5}\right) \cdot \left(\frac{1}{5}\right)$

(c) $\left(\frac{1}{5}\right) \cdot \left(\frac{1}{5}\right) \cdot \left(\frac{1}{5}\right)$ ✓

Reading Numbers in Exponential Notation

Below are some powers of 3 and how they are read in exponential notation.

$$
\begin{array}{rcll}
3 & = & 3^1 & \text{“Three or three to the first power.”} \\
3 \cdot 3 & = & 3^2 & \text{“Three to the second power or three } \textbf{squared}\text{.”} \\
3 \cdot 3 \cdot 3 & = & 3^3 & \text{“Three to the third power or three } \textbf{cubed}\text{.”} \\
3 \cdot 3 \cdot 3 \cdot 3 & = & 3^4 & \text{“Three to the fourth power.”} \\
3 \cdot 3 \cdot 3 \cdot 3 \cdot 3 & = & 3^5 & \text{“Three to the fifth power.”} \\
 & \vdots & &
\end{array}
$$

Question 8 *Write the expression below in words.*

$$100^3$$

Multiple Choice:

(a) *One hundred three*

(b) *Three hundred*

(c) *One hundred times three*

(d) *One hundred cubed* ✓

Question 9 *Write the expression below in words.*

$$10^2 \cdot 3^6$$

Multiple Choice:

(a) *Ten times 2 times 3 times 6*

(b) *Ten squared times three to the sixth power* ✓

(c) *Thirty to the twelfth power*

(d) *Ten to the sixth power times three squared*

Positive Powers of 10

A **positive power of 10** is a power of 10 in which the exponent is a **positive integer**: $\{1, 2, 3, 4, \ldots\}$. Below are the first three positive powers of 10. **Notice that the exponent in each power of 10 matches the number of zeros in the result**. In general, if n is a positive integer, then the value of 10^n is the number 1 followed by n zeros. For example, below you can see that 10^3 equals the number 1 followed by 3 zeros.

$$\begin{aligned} 10^1 &= 10 \ \rightarrow 1 \text{ zero} \\ 10^2 &= 100 \ \rightarrow 2 \text{ zeros} \\ 10^3 &= 1,000 \ \rightarrow 3 \text{ zeros} \end{aligned}$$

Question 10 *Write the value of* 10^4.

Multiple Choice:

(a) 40

(b) $10,000$ ✓

(c) $1,000$

(d) $40,000$

Question 11 *Write the value of* 10^6. ***Do not include commas in your answer.***

$$10^6 = 1000000$$

What number is this?

Multiple Choice:

(a) *One hundred thousand*

(b) *One million* ✓

(c) *One billion*

(d) *Ten thousand*

Question 12 *A greedy CEO demanded a pay package worth \$1 trillion* $= 10^{12}$. *Write the value of* 10^{12}.

Multiple Choice:

(a) 120

(b) $10,000,000,000$

(c) $1,000,000,000$

(d) $1,000,000,000,000$ ✓

Extra Insight: The Name Behind the Search

A **googol** is 10^{100}, the number 1 followed by 100 zeros! In 1920, mathematician Edward Kasner was writing a book called *Mathematics and Imagination* and wanted to give a name to 10^{100}. He asked his 9-year-old nephew, Milton Sirotta, for a suggestion and Milton came up with the word googol. Decades later, it inspired the name of one of the largest and well known tech companies: Google.

Multiplying and Dividing by Positive Powers of 10

When we multiply or divide a number by a positive power of 10, there is a relationship between the movement of the decimal point and the number of zeros in the power of 10. **Once you notice this relationship, you won't need to perform multiplication or long division to find the answers to these types of problems**.

Multiplying by Positive Powers of 10

Look for the relationship between the number of zeros in the power of 10 and the movement of the decimal point in the multiplication problems shown below.

$0.625(10)$	$=$	6.25	The decimal point moved 1 place to the right.
$0.625(100)$	$=$	62.5	The decimal point moved 2 places to the right.
$0.625(1000)$	$=$	625	The decimal point moved 3 places to the right.

$5(10)$	$=$	50	The decimal point moved 1 place to the right.
$5(100)$	$=$	500	The decimal point moved 2 places to the right.
$5(1000)$	$=$	5000	The decimal point moved 3 places to the right.

Question 13 *Write a general rule for multiplying a number by a positive power of 10.*

To multiply a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the (left/ right ✓) that many places.

Question 14 *Use the rule from the previous problem to multiply.*

$$2.8(10) = 28$$

$$3.0525(1000) = 3052.5$$

$$620(100) = 62000$$

Dividing by Positive Powers of 10

Look for the relationship between the number of zeros in the power of 10 and the movement of the decimal point in the division problems shown below.

$$\frac{43.2}{10} = 4.32$$ The decimal point moved 1 place to the left.

$$\frac{43.2}{100} = 0.432$$ The decimal point moved 2 places to the left.

$$\frac{43.2}{1000} = 0.0432$$ The decimal point moved 3 places to the left.

$$\frac{3}{10} = 0.3$$ The decimal point moved 1 place to the left.

$$\frac{3}{100} = 0.03$$ The decimal point moved 2 places to the left.

$$\frac{3}{1000} = 0.003$$ The decimal point moved 3 places to the left.

Question 15 *Write a general rule for dividing a number by a positive power of 10.*

To divide a number by a positive power of 10, count the number of zeros in the power of 10, then move the decimal point to the (left ✓/ right) that many places.

Question 16 *Use the rule from the previous problem to divide.* ***Do not include commas in your answer****.*

$$\frac{63}{1000} = 0.063$$

$$\frac{35.5}{10} = 3.55$$

$$\frac{1}{100} = 0.01$$